

Estimation theory:

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estimation: what is the value of an unknown quantity?

eg Toss a coin of unknown bias n times, with results $y_1, \dots, y_n$. Then what is the value of the bias of the coin?

(Detection Version: Coin with bias p_a or p_b . Toss coin ~~many times~~ n times, and guess whether bias is p_a or p_b).

defn: A point estimator (estimator) is any function $f(x_1, \dots, x_n)$ of n samples, with values in Θ of parameters.
a set

defn: A parameter estimation / point estimation problem is to observe $(x_1, \dots, x_n) \sim p_\theta$, $\theta \in \Theta$ (Θ is unknown), and estimate $\theta \in \Theta$.

eg (Sample mean
sample median)

Methods of designing estimators -

1) Method of moments estimators:

$$(x_1, \dots, x_n) \stackrel{\text{iid}}{\sim} p_\theta, \quad x_i \in \mathbb{R}$$

$$\theta = (\theta_1, \dots, \theta_k) \quad \theta_i \in \mathbb{R}$$

$$\mu_1(\theta) = E_{x \sim p_\theta}[x] = E_\theta[x] = \int x p_\theta(x) dx$$

$$\approx \frac{1}{n} \sum_{i=1}^n X_i \quad - (1)$$

$$\mu_2(\theta) = E_\theta[x^2] \approx \frac{1}{n} \sum_{i=1}^n X_i^2 \quad - (2)$$

⋮

$$\mu_k(\theta) = E_\theta[x^k] \approx \frac{1}{n} \sum_{i=1}^n X_i^k \quad - (k)$$

output $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_k)$
that solves eqn (1) ... (k)

eg Normal method of moments estimators:

$$X_1, \dots, X_n \sim N(\mu, \sigma^2)$$

both μ and σ^2
are unknown.

Solve, $\theta = (\theta_1, \theta_2) = (\mu, \sigma^2)$

$$\hat{\theta}_1 = \frac{1}{n} \sum_{i=1}^n X_i \quad - (i)$$

$$(\hat{\theta}_1)^2 + \hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^n X_i^2 \quad - (ii)$$

hence,

$$\left. \begin{aligned} \hat{\theta}_1 &= \frac{1}{n} \sum_{i=1}^n X_i \\ \hat{\theta}_2 &= \frac{1}{n} \sum_{i=1}^n X_i^2 - \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2 \end{aligned} \right\}$$

$\hat{\mu}$ points to $\hat{\theta}_1$
 $\hat{\sigma}^2$ points to $\hat{\theta}_2$

$$\underline{\underline{\frac{1}{n} \sum_{i=1}^n X_i^2 - \hat{\theta}_1^2}}$$

Problem: observe

$$\mathbf{X} \sim P_{\theta}$$

(X could be $X_1, \dots, X_n$)

Goal: find $g(\theta)$ where $g(\cdot)$ is some known functions —

1) Method of Moments :

eg Method of moments for Binomial

$$X_1 \dots X_n \stackrel{iid}{\sim} \text{Bin}(\overset{(m)}{\theta_1}, \overset{(p)}{\theta_2})$$

$$\in \mathbb{N} \quad \in [0, 1]$$

$$P(X_i = k) = \binom{m}{k} p^k (1-p)^{m-k}$$

$$\rightarrow E_{\theta}[x] = mp = \frac{1}{n} \sum_{i=1}^n X_i$$

$$E_{\theta}[x^2] = mp(1-p) + (mp)^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$$

Solving the two eqn we get,

$$\hat{m} = \left(\frac{\bar{X}^2}{\left(\bar{X} - \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \right)} \right), \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\hat{p} = \left(\frac{\bar{X}}{\hat{m}} \right)$$

Note $\rightarrow$

(i) The denominator

$$\bar{x} - \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \equiv -ve$$

and hence $\hat{m}$ and $\hat{\beta}$ also,

(ii) $\hat{m}$ also need not be integer valued.

2) Maximum Likelihood Estimators:

- Most popular principle behind estimator design

Idea: $X \sim P_\theta$, $\theta \in \Theta$

$$\rightarrow \text{ML estimate of } \theta : \underset{\theta \in \Theta}{\operatorname{argmax}} P_\theta(x)$$
$$\hat{\theta}_{ML}(x) =$$

$\rightarrow$ The ML estimate always lies in Θ .

$\rightarrow$ The ML estimate is solution to the constrained optimization problem.

eg Gaussian ML estimate -

$x_1 \dots x_n \stackrel{iid}{\sim}$

$N(\mu, \sigma^2)$

$\theta \equiv (\theta_1, \theta_2)$ - unknown

$$\log P_\theta(x) = \log \prod_{i=1}^n \frac{1}{\sqrt{2\pi\theta_2}} e^{-\frac{(x_i - \theta_1)^2}{2\theta_2}}$$

$$\log P_\theta(x) = \left[\sum_{i=1}^n \left(\frac{1}{2} \log \left(\frac{1}{2\pi\theta_2} \right) - \frac{1}{2\theta_2} (x_i - \theta_1)^2 \right) \right]$$

$$\hat{\theta}_{ML} = \arg \max_{\substack{\theta_1 \in \mathbb{R} \\ \theta_2 > 0}} \log P_{\theta}(x) \quad (53)$$

$$\begin{aligned} \hat{\theta}_{ML} &= \arg \max_{\theta_1, \theta_2} \sum_{i=1}^N \left[\frac{1}{2} \log \left(\frac{1}{2\pi\theta_2} \right) - \frac{1}{2\theta_2} (x_i - \theta_1)^2 \right] \\ &= \arg \max_{\theta_1, \theta_2} \sum_{i=1}^N \left[\frac{1}{2} \log \left(\frac{1}{\theta_2} \right) - \frac{1}{2\theta_2} (x_i - \theta_1)^2 \right] \end{aligned}$$

$$(i) \quad \frac{\partial}{\partial \theta_1} (\cdot) = \sum_{i=1}^N \left[-\frac{1}{2\theta_2} 2(x_i - \theta_1) \times (-1) \right] \stackrel{\text{Set}}{=} 0$$

$$\Rightarrow \sum_{i=1}^N \frac{1}{\theta_2} (x_i - \theta_1) \stackrel{\text{Set}}{=} 0$$

$$(ii) \quad \frac{\partial}{\partial \theta_2} (\cdot) = -\frac{n}{2\theta_2} + \frac{1}{2\theta_2^2} \sum_i (x_i - \theta_1)^2 \stackrel{\text{Set}}{=} 0$$

after solving we get -

$$\hat{\theta}_1 = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\theta}_1)^2$$

$$\frac{1}{n} \sum_{i=1}^n x_i^2 + \frac{1}{n} \sum_{i=1}^n \hat{\theta}_1^2 - \frac{2}{n} \sum_{i=1}^n x_i \hat{\theta}_1$$

$$\frac{1}{n} \sum_{i=1}^n x_i^2 + \hat{\theta}_1^2 - 2\hat{\theta}_1^2$$

$$\frac{1}{n} \sum_{i=1}^n x_i^2 - \hat{\theta}_1^2$$

→ 2nd order derivatives must be checked -

e.g. Bernoulli (θ) :

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$

$$\hat{\theta}_{ML} = \arg \max_{\theta \in [0,1]} P_{\theta}(x_1, \dots, x_n)$$

$$\hat{\theta}_{ML} = \arg \max_{\theta \in [0,1]} \left(\theta^{\sum_{i=1}^n x_i} (1-\theta)^{n - \sum_{i=1}^n x_i} \right)$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i) \quad \left. \vphantom{\frac{1}{n} \sum_{i=1}^n (x_i)} \right\} \underline{\text{Checked}}$$

III) Bayes (or Bayesian) estimation

$$X \sim P_{\theta}, \theta \in \Theta$$

Idea: (i) Assume a prior distribution over Θ .

(ii) Compute posterior distribution over Θ , after observing X .

(iii) Output a suitable function of the posterior distribution.

eg Bayes estimation
for Ber' distribution —

$$\begin{aligned} X_1 \dots X_n &\stackrel{\text{iid}}{\sim} \text{Ber}(\theta) \\ 0 &\leq \theta \leq 1 \\ \Theta &\in [0,1] \end{aligned}$$

Prior on Θ : Unif $[0,1]$

$$\pi(\theta) = 1_{\{0 \leq \theta \leq 1\}}$$

→ posterior density after observing $X \triangleq (x_1, \dots, x_n)$

$$\pi(\theta | x_1, \dots, x_n) =$$

$$\frac{\pi(x_1=x_1, \dots, x_n=x_n | \theta) \pi(\theta)}{\int_0^1 \pi(x_1=x_1, \dots, x_n=x_n | \theta') \pi(\theta') d\theta'}$$

$$= \frac{\theta^{\sum x_i} (1-\theta)^{n-\sum x_i} \mathbb{1}_{\{0 \leq \theta \leq 1\}}}{\int_0^1 \theta^{\sum x_i} (1-\theta')^{n-\sum x_i} d\theta'} \quad (54)$$

$$= \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1} \mathbb{1}_{\{0 \leq \theta \leq 1\}}}{C}$$

, where α, β and C do not depend on θ .

$$= \text{Beta}(\alpha, \beta)$$

with $\alpha = 1 + \sum_{i=1}^n x_i$
 $\beta = 1 + n - \sum_{i=1}^n x_i$
 $\alpha, \beta > 0$

→ One can output an estimate:

$$\begin{aligned} \text{(i)} \quad \text{Mean}(\text{Beta}(\alpha, \beta)) &= \frac{\alpha}{\alpha + \beta} \\ &= \frac{1 + \sum_{i=1}^n x_i}{2 + n} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \text{Mode}(\text{Beta}(\alpha, \beta)) &= \underset{\theta \in [0,1]}{\text{argmax}} f(\theta) \\ &= \frac{\alpha-1}{\alpha+\beta-2} \\ &= \frac{\sum_{i=1}^n x_i}{n} \end{aligned}$$

$$\text{(iii)} \quad \text{Median}(\text{Beta}(\alpha, \beta)) \approx \frac{\alpha - 1/3}{\alpha + \beta - 2/3}$$

Remark : ① $\text{unif}(0,1) = \text{Beta}(1,1)$

② If $\pi(\theta) = \text{Beta}(a,b)$

$$\pi(\theta|x) = \text{Beta}(\alpha, \beta)$$

$$\left. \begin{aligned} \alpha &= a + \sum_i x_i \\ \beta &= b + n - \sum_i x_i \end{aligned} \right\}$$

③ The family Beta distribution is a conjugate prior family for the family of Bernoulli distribution.

e.g. Bayesian estimation for the Normal distribution:

$$x_1, \dots, x_n \stackrel{\text{iid}}{\sim} N(\theta, 1) \quad \theta \in \mathbb{R} = \Theta$$

$$\text{prior distribution on } \mathbb{R}: N(\mu, \tau^2)$$

It follows that:

$$P(\theta | x) = \frac{P(x | \theta) P(\theta)}{\int}$$

$$= N\left(\frac{\frac{\tau^2}{n} \cdot \bar{x} + \frac{1}{n} \cdot \mu}{\frac{\tau^2}{n} + 1}, \frac{\tau^2}{\frac{\tau^2}{n} + 1}\right)$$

i.e. Normal is a conjugate prior for Normal w/ fixed Variance.

→ Using the mean / median / mode
gives the Bayesian estimator —

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$$\bar{x} = \frac{1}{n} \sum_i x_i$$

$$\text{estimator: } \left(\left(\frac{z^2}{\frac{1}{n} + z^2} \right) \bar{x} + \left(\frac{\frac{1}{n}}{\frac{1}{n} + z^2} \right) \mu \right)$$